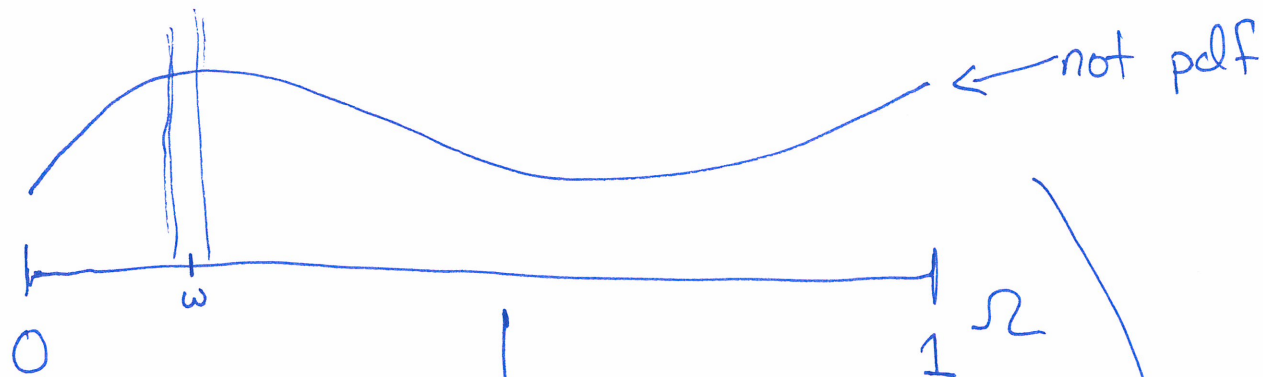
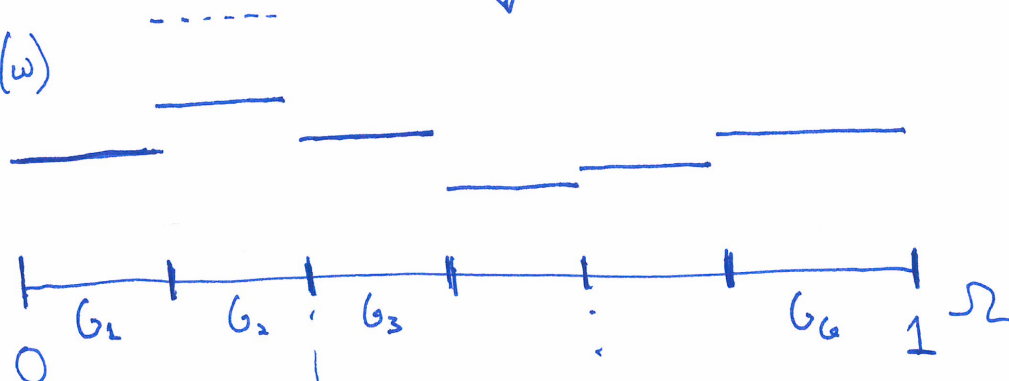


$X(\omega)$

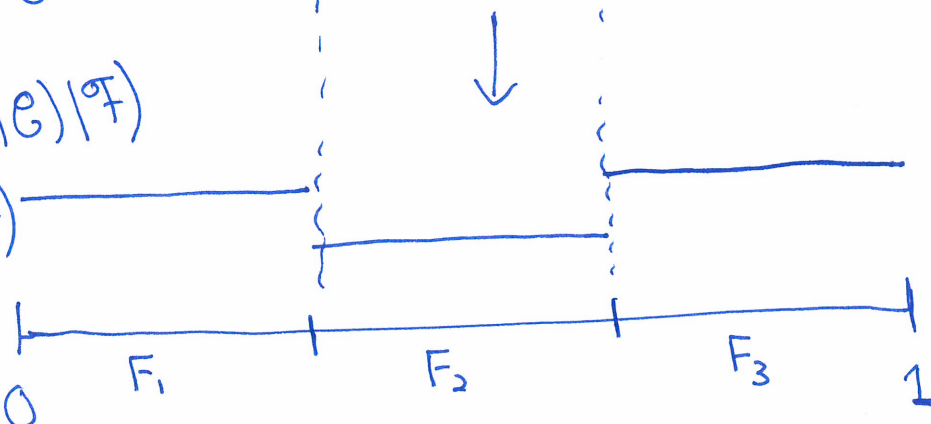


$E(X|\mathcal{G})(\omega)$



$E(E(X|\mathcal{G})|\mathcal{F})$

$=E(X|\mathcal{F})$



$$\mathcal{F} \subseteq \mathcal{G}$$

Think about $\mathcal{F} = \{\emptyset, \Omega\}$. Then $E(X|\mathcal{F}) = E(X)$

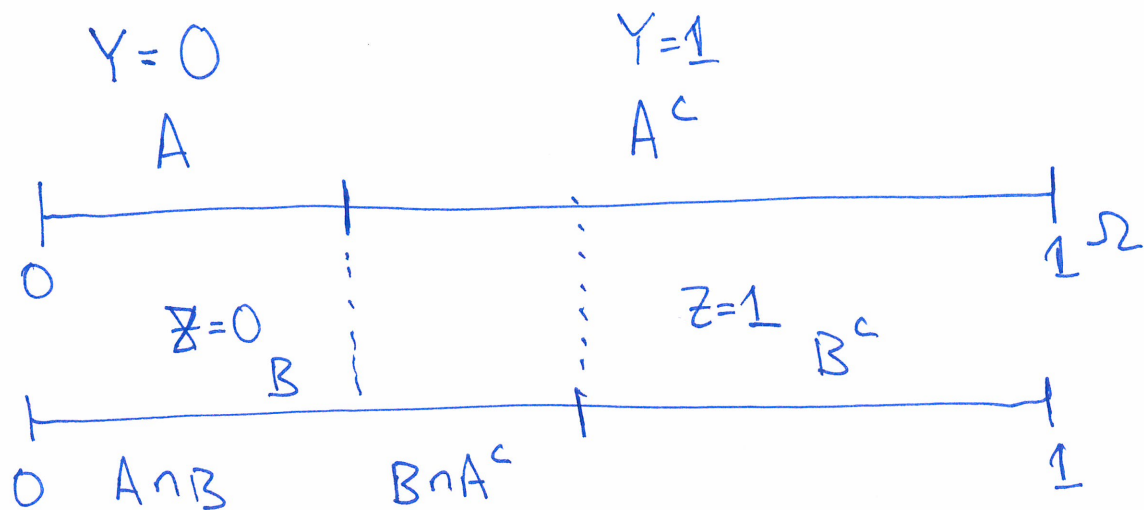
$$E(X|Y, Z)$$

$$\sigma(Y, Z) \neq \sigma(Y) \cup \sigma(Z)$$

$$\mathcal{F} = \{\emptyset, \Omega, A, A^c\}, \quad \mathcal{G} = \{\emptyset, \Omega, B, B^c\}$$

$$\mathcal{F} \cup \mathcal{G} = \{\emptyset, \Omega, A, A^c, B, B^c\}$$

Does not contain $A \cup B$



The Delta Method

A formal statement of the Delta Method is as follows:

Assume that $Y_1, Y_2, \dots$ are such that

$$\frac{Y_n - \mu}{\sigma_n} \xrightarrow{D} Z$$

where $Z \sim N(0, 1)$, μ is a constant, and σ_n is such that $\sigma_n > 0$ and $\sigma_n \downarrow 0$ as $n \rightarrow \infty$. Then, assuming $g(\cdot)$ is a function that is differentiable at μ and $g'(\mu) \neq 0$, it holds that

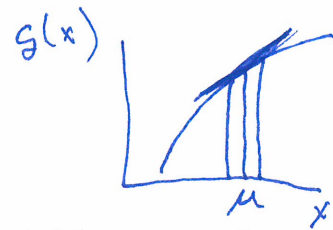
$$\frac{g(Y_n) - g(\mu)}{|g'(\mu)|\sigma_n} \xrightarrow{D} Z.$$

A more heuristic, but useful, version is as follows:

If Y_n is approximately $N(\mu, \sigma_n^2)$ and $\sigma_n^2 \downarrow 0$ as $n \rightarrow \infty$, then $g(Y_n)$ is approximately

$$N(g(\mu), (g'(\mu))^2 \sigma_n^2)$$

assuming that $g(\cdot)$ is a function that is differentiable at μ and $g'(\mu) \neq 0$.



This is often (but not always) applied with $Y_n = \bar{X}_n$ and $\sigma_n^2 = \sigma^2/n$ where $\sigma^2 > 0$.

$$V(\bar{X}_n) = \frac{\sigma^2}{n}$$
$$\sigma^2 = V(X_i)$$

Exercise: Suppose that $X_1, X_2, \dots, X_{50}$ are i.i.d. random variables with the Exponential($\lambda = 2$) distribution. One may hope that $1/\bar{X}$ is a good estimator of λ (imagining a case where λ is not known). What is the probability that $1/\bar{X}$ is within 0.1 of λ ?

$$E(X_i) = 1/\lambda, \quad \lambda = 1/E(X_i) \quad \bar{X} \approx E(X_i) \quad \lambda \approx 1/\bar{X}$$

$$\mu = E(X_i) = 1/2, \quad \sigma^2 = V(X_i) = 1/4$$

$$\text{By CLT, } \bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(\frac{1}{2}, \frac{1}{200}\right)$$

$$\text{For now, be more general, say that} \\ \bar{X} \approx N\left(\frac{1}{\lambda}, \frac{1}{\lambda^2 n}\right)$$

Want the approx. dist. of $1/\bar{X}$.

$$\text{Use the } \Delta\text{-method with } g(\bar{X}) = 1/\bar{X} \\ \text{i.e. } g(x) = 1/x$$

$$\text{So, } g'(x) = -1/x^2, \quad g'(\mu) = -\lambda^2 \neq 0$$

$$\text{By the } \Delta\text{-method, } g(\bar{X}) = 1/\bar{X} \text{ is approx.} \\ \text{Normal with mean } g(\mu) = \lambda \text{ and} \\ \text{variance } \left(g'(\mu)\right)^2 \frac{\sigma^2}{n} = \lambda^4 \left(\frac{1}{\lambda^2 n}\right) = \lambda^2/n$$

So, by Δ -method,

$$1/\bar{X} \approx N(\lambda, \lambda^2/n)$$

In fact, $\bar{X} \sim \text{Gamma}(n, n\lambda)$

and $1/\bar{X} \sim \text{Inverse Gamma}(n, n\lambda)$

If $\lambda=2$, $n=50$, $\bar{X} \approx N(2, 4/50)$

$P(\bar{X} \text{ is within } 0.1 \text{ of } 2)$

$$= P(1.9 \leq \bar{X} \leq 2.1)$$

$$= F(2.1) - F(1.9)$$

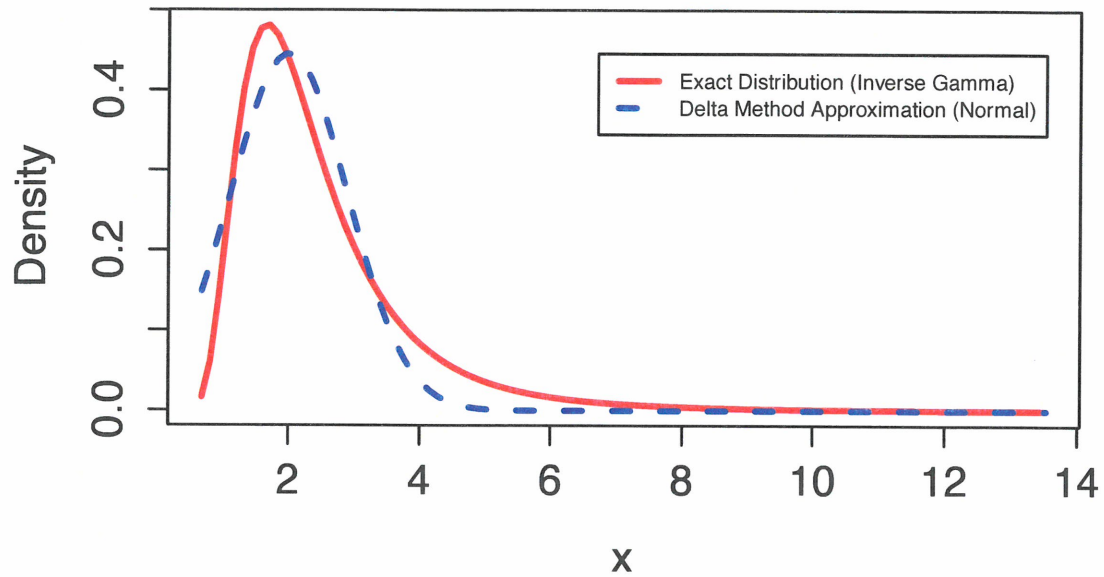
where F is the CDF for
normal $(2, 4/50)$ distribution.

$\hat{\theta}$ is the MLE for θ

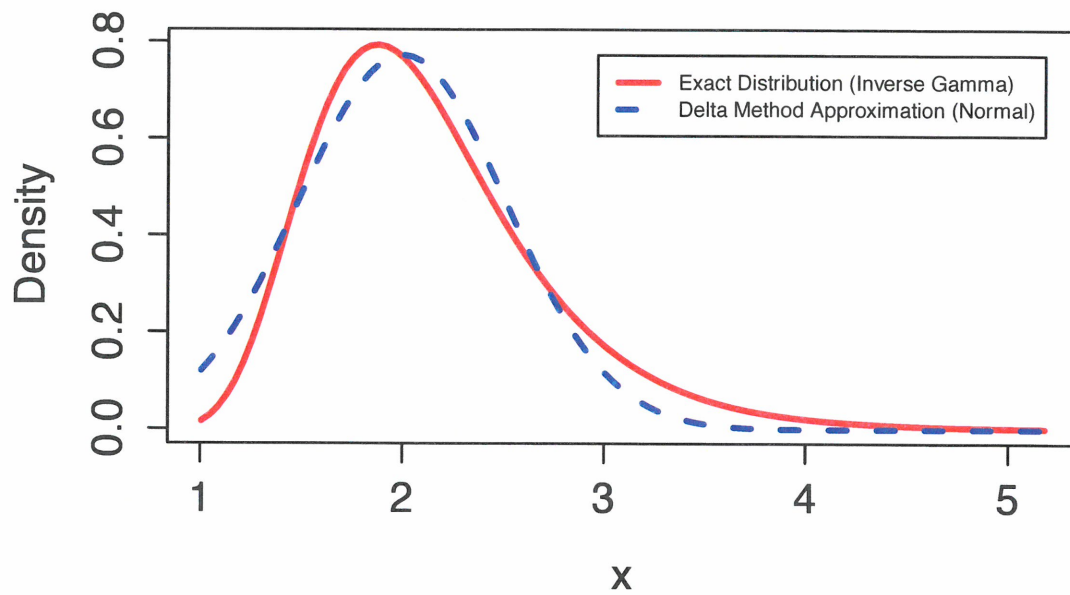
$$\hat{\theta} \approx N(\theta_0, V(\hat{\theta}))$$

Δ -method says that $g(\hat{\theta})$ is also
approx. normal

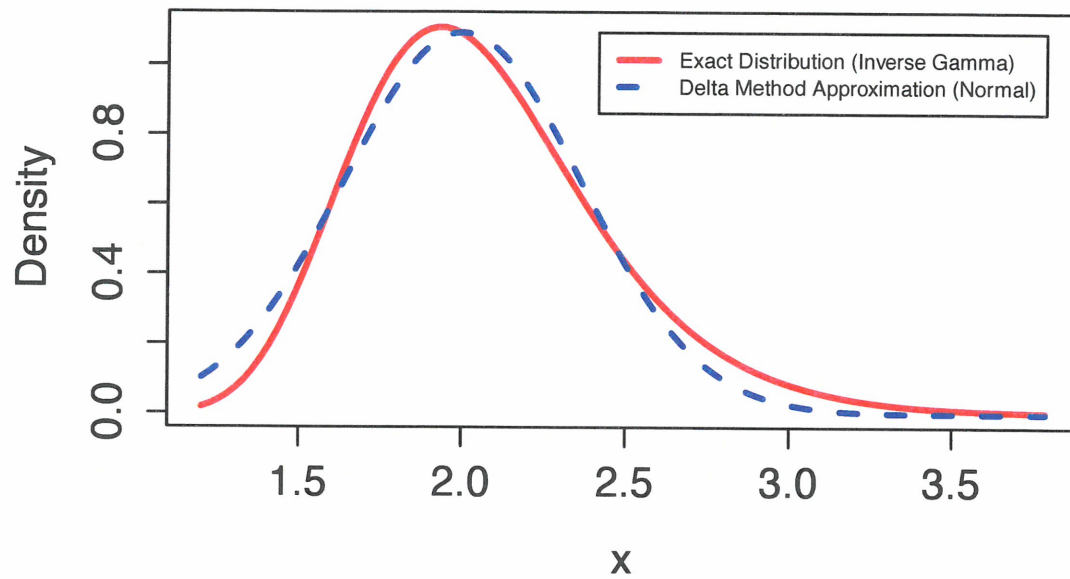
With $n = 5$



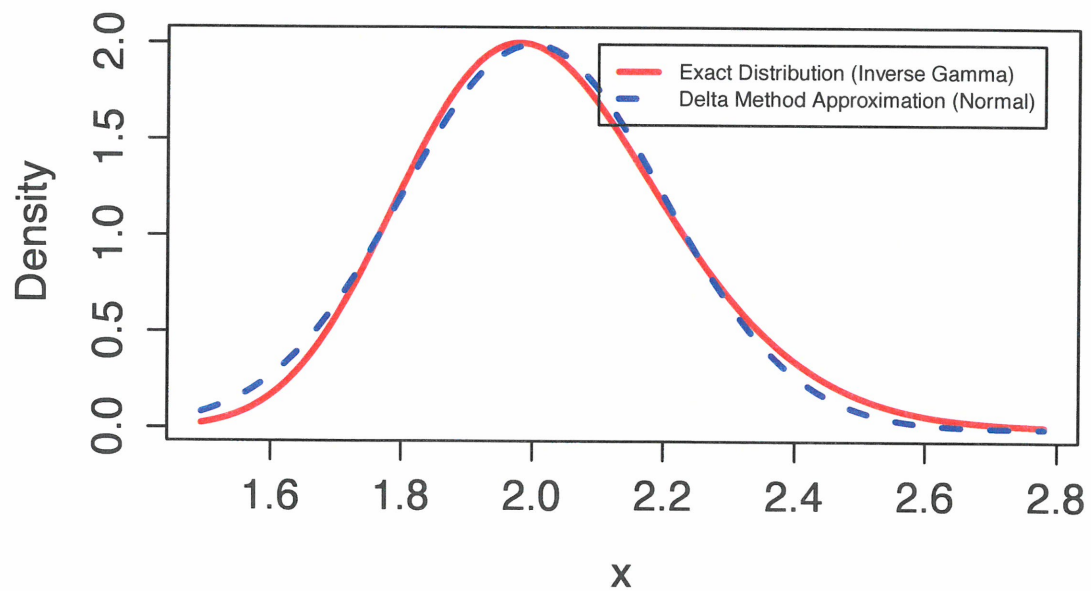
With $n = 15$



With $n = 30$



With $n = 100$



¹ Part 6: Four Classic Results

- ² This brief Part presents important results that relate some of the proba-
³ bility material discussed to this point to important concepts in **statistical**
⁴ **inference**.

$k=1, 2, \dots$
1 The **Chi-squared distribution with k degrees of freedom** is a special case
2 of the Gamma distribution. In particular, if X has the $\text{Gamma}(k/2, 1/2)$
3 distribution, then X has the Chi-square distribution with k degrees of free-
4 dom.

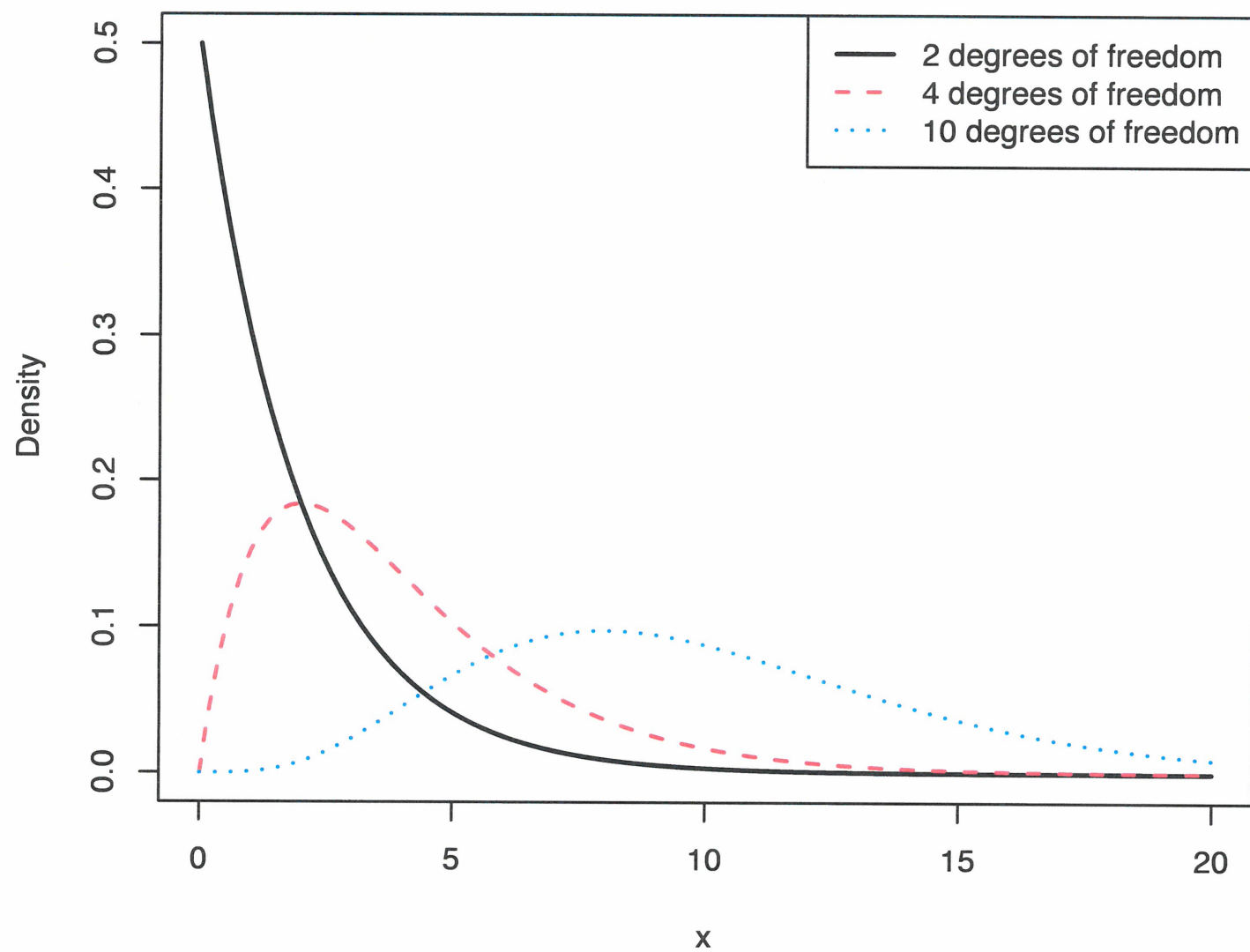
5 Also, if $Z_1, Z_2, \dots, Z_k$ are i.i.d. standard normal random variables, then

6
$$X = \sum_{i=1}^k Z_i^2$$

7 has the Chi-squared distribution with k degrees of freedom.

8 If X has this distribution, then $E(X) = k$ and $V(X) = 2k$.

9 Plots of pdfs from the Chi-squared distribution are shown on the following
10 slide.



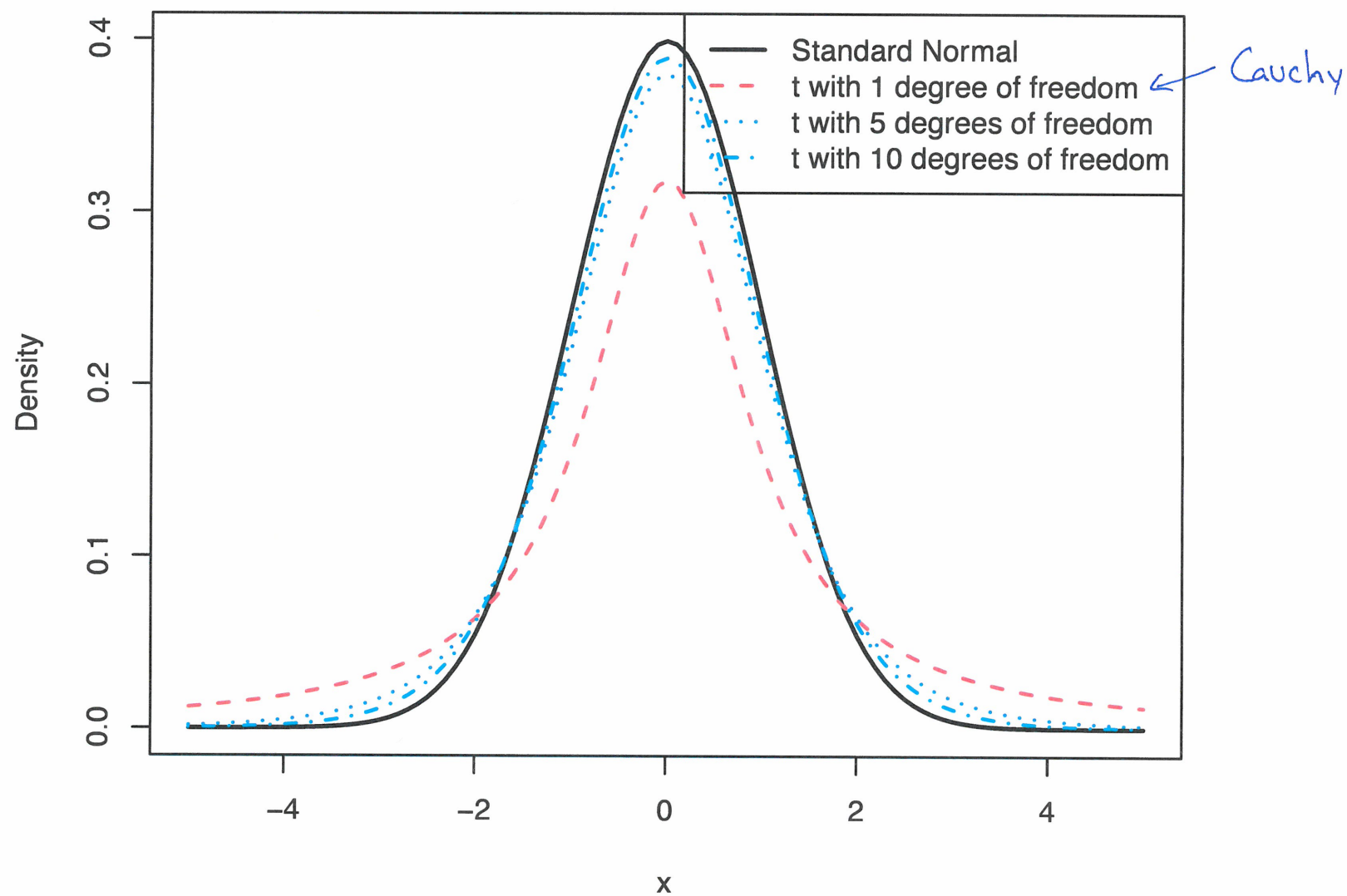
- 1 The t -distribution with ν degrees of freedom is defined to be a random
2 variable T that can be constructed as

3
$$T = \frac{Z}{\sqrt{X/\nu}}$$

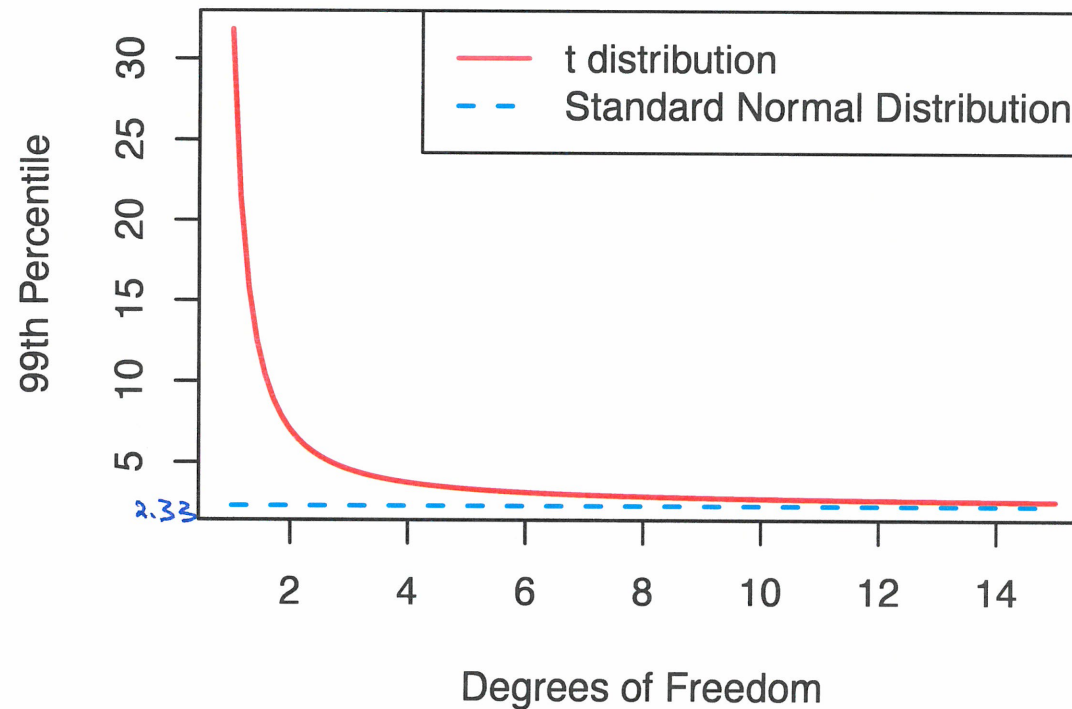
- 4 where Z is a random variable with the standard normal distribution, and
5 X is a random variable with the Chi-squared distribution with ν degrees
6 of freedom. Further, Z and X are assumed independent.

- 7 This distribution has a “normal-shaped” pdf, but has heavier tails than
8 the normal distribution. As ν increases, the distribution converges to the
9 standard normal distribution.

- 10 It holds that $E(T) = 0$ when $\nu \geq 2$ and that $V(T) = \nu/(\nu - 2)$ when $\nu \geq 3$.



- 1 The figure below makes clearer how much additional probability is in the
2 tails of the t -distribution compared to the standard normal. For $\nu = 1$,
3 1% of the observations will be larger than 30. The 99th percentile of the
4 standard normal distribution is 2.33.



1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. t -distributed with ν degree
2 of freedom. What is the (approximate) distribution of the sample mean of
3 the X_i ?

4 When $\nu=1$ or $\nu=2$, $V(X_i)$ is undefined,
5 so CLT does not hold

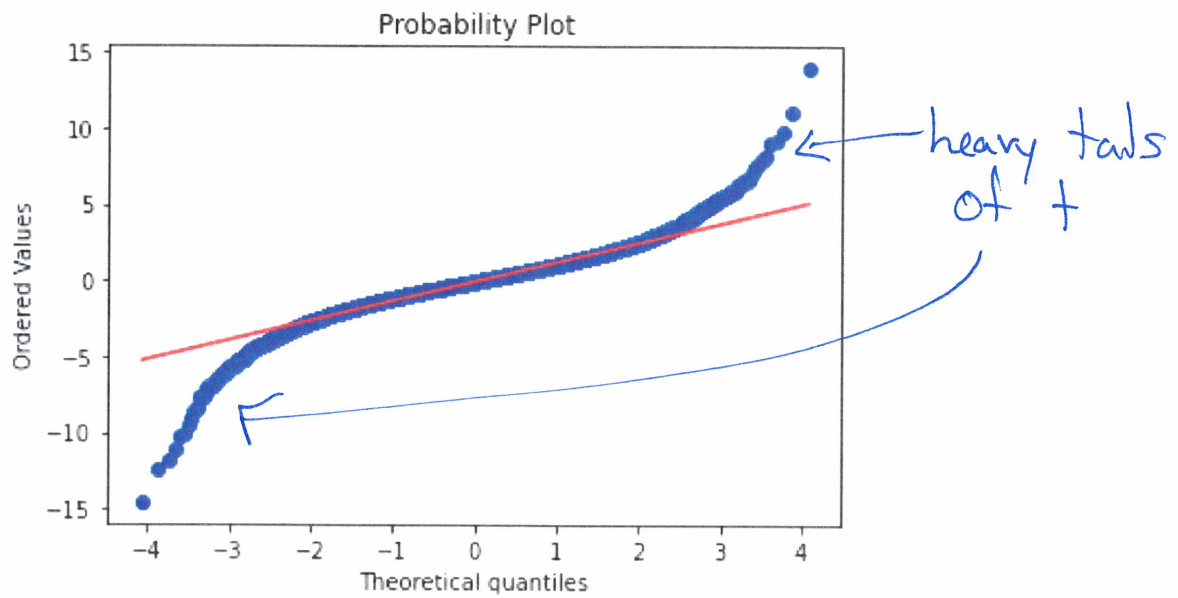
6
7 For $\nu > 2$, CLT says that $\bar{X} \approx \text{Normal}$
8
9
10
11
12

```
In [1]: from scipy.stats import t, probplot
import numpy as np
import matplotlib.pyplot as plt
```

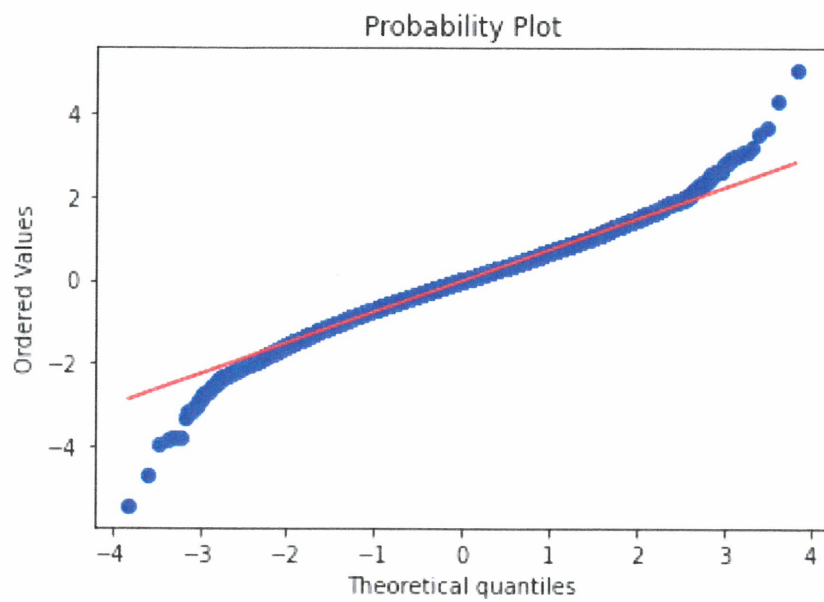
```
In [14]: n = 3
reps = 10000
nu = 5
```

```
In [15]: draws = t.rvs(nu, size=[reps,n])
```

```
In [21]: res = probplot(np.ndarray.flatten(draws), plot=plt)
```



```
In [16]: res = probplot(np.mean(draws,1), plot=plt)
```



In []:

1 Now we can state four **classic results** that hold in the i.i.d. Normal case:

2 Assuming that $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$, then

3 1. $\bar{X}$ and s^2 are independent.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

"sample variance"

4 2. The quantity

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

6 has the standard normal distribution.

7 3. The quantity

$$\frac{(n-1)s^2}{\sigma^2}$$

9 has the Chi-squared distribution with $n-1$ degrees of freedom.

10 4. The quantity

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}$$

12 has the t -distribution with $n-1$ degrees of freedom.

Proofs of the Classic Results

Throughout assume that $X_1, X_2, \dots, X_n$ are i.i.d. from the normal distribution with mean μ and variance σ^2 .

Claim 1: $\bar{X}$ and S^2 are independent.

Proof: First, note that since $X_1, X_2, \dots, X_n$ are independent and normal, then the random vector $\mathbf{X}$ consisting of $X_1, X_2, \dots, X_n$ is multivariate normal with covariance matrix $\Sigma = \sigma^2 \mathbf{I}$. (To see this, derive the joint pdf of $\mathbf{X}$ by taking the product of the pdfs of $X_1, X_2, \dots, X_n$ and note that its form matches the multivariate normal with covariance matrix Σ .)

Fix any $i \in \{1, 2, \dots, n\}$. We see that we can write

$$\begin{bmatrix} \bar{X} \\ (X_i - \bar{X}) \end{bmatrix} = \mathbf{A}_i \mathbf{X}$$

with

$$\mathbf{A}_i = \begin{bmatrix} \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \dots & \frac{1}{n} \\ -\frac{1}{n} & \dots & -\frac{1}{n} & \underbrace{1 - \frac{1}{n}} & -\frac{1}{n} & \dots & -\frac{1}{n} \end{bmatrix}$$

where the $1 - 1/n$ entry in the second row of $\mathbf{A}_i$ is in the i^{th} column. This implies that

$$\begin{bmatrix} \bar{X} \\ (X_i - \bar{X}) \end{bmatrix}$$

is bivariate normal with covariance matrix $\mathbf{A}_i \Sigma \mathbf{A}_i^T = \sigma^2 \mathbf{A}_i \mathbf{A}_i^T$. Finally, the off-diagonal entry of $\sigma^2 \mathbf{A}_i \mathbf{A}_i^T$ is

$$\sigma^2 \left[\left(\frac{1}{n} \right) \left(1 - \frac{1}{n} \right) + \sum_{j \neq i} \left(\frac{1}{n} \right) \left(-\frac{1}{n} \right) \right] = 0.$$

Of course, this off-diagonal element is the covariance between $\bar{X}$ and $(X_i - \bar{X})$. The conclusion is that $\bar{X}$ and $(X_i - \bar{X})$ are uncorrelated and hence independent.

Next, note that the sample variance S^2 can be written as a function of $(X_i - \bar{X})$ for $i = 1, 2, \dots, n$:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Hence, S^2 is a function of random variables, each of which is independent of $\bar{X}$. This implies that $\bar{X}$ is independent of S^2 . $\square$

Claim 2: The quantity

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has the standard normal distribution.

Proof: This is a direct consequence of results already stated in Probability. $\square$

Claim 3: The quantity

$$\frac{(n-1)S^2}{\sigma^2}$$

has the chi-squared distribution with $n-1$ degrees of freedom.

Proof: To start, note that

$$\begin{aligned} \sum_{i=1}^n \underbrace{\left(\frac{X_i - \mu}{\sigma}\right)^2}_{\text{std. normal}} &= \sum_{i=1}^n \left(\frac{X_i - \bar{X} + \bar{X} - \mu}{\sigma}\right)^2 \\ &= \frac{1}{\sigma^2} \sum_{i=1}^n \left[(X_i - \bar{X})^2 + 2(X_i - \bar{X})(\bar{X} - \mu) + (\bar{X} - \mu)^2 \right] \\ &= \frac{1}{\sigma^2} \left[\sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2 \right] \quad \text{since } \sum_i (X_i - \bar{X}) = 0 \\ &= \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 + \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right)^2 \\ &= W_2 + W_3 \end{aligned}$$

where

$$W_2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma}\right)^2 \quad \text{and} \quad W_3 = \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right)^2.$$

Also define

$$W_1 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma}\right)^2$$

so that we have shown $W_1 = W_2 + W_3$.

We know that W_1 has the chi-squared distribution with n degrees of freedom and that W_3 has the chi-squared distribution with a single degree of freedom. Note that the only random variable W_2 depends on is S^2 and the only random variable W_3 depends on is $\bar{X}$.

Since S^2 and $\bar{X}$ are independent, it follows that W_2 and W_3 are independent. Hence,

$$M_{W_1}(t) = M_{W_2}(t)M_{W_3}(t)$$

and

$$\begin{aligned} M_{W_2}(t) &= M_{W_1}(t) / M_{W_3}(t) \\ &= \left(\frac{1}{1-2t} \right)^{n/2} / \left(\frac{1}{1-2t} \right)^{1/2} \\ &= \left(\frac{1}{1-2t} \right)^{\left(\frac{n-1}{2} \right)}. \end{aligned}$$

So, by the uniqueness of the MGF, W_2 has the chi-squared distribution with $n - 1$ degrees of freedom. Finally,

$$W_2 = \sum_{i=1}^n \left(\frac{X_i - \bar{X}}{\sigma} \right)^2 = \frac{(n-1)S^2}{\sigma^2}. \quad \square$$

Claim 4: The quantity

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}$$

has the t -distribution with $n - 1$ degrees of freedom.

Proof: Start by defining random variables

$$Z = \left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \right) \quad \text{and} \quad V = \frac{(n-1)S^2}{\sigma^2}.$$

Based on the above results, we know that Z has the standard normal distribution and V has the chi-squared distribution with $n - 1$ degrees of freedom. We also know that Z and V are independent.

It is not difficult to see that

$$\frac{\bar{X} - \mu}{s/\sqrt{n}} = \frac{Z}{\sqrt{V/(n-1)}}$$

and thus by the definition of the t -distribution, the claim follows. $\square$

- 1 **Exercise:** What are the expected value and variance of s^2 ? What is the
2 practical significance of these?

$$\begin{aligned} E(s^2) &= E\left(\left(\frac{\sigma^2}{n-1}\right)\left(\frac{n-1}{\sigma^2}\right)S^2\right) = \left(\frac{\sigma^2}{n-1}\right)E\left(\left(\frac{n-1}{\sigma^2}\right)S^2\right) \\ &= \left(\frac{\sigma^2}{n-1}\right)(n-1) = \sigma^2 \end{aligned}$$

3
4
5
6 " s^2 is an unbiased estimator for σ^2 "

$$\begin{aligned} V(s^2) &= V\left(\left(\frac{\sigma^2}{n-1}\right)\left(\frac{n-1}{\sigma^2}\right)S^2\right) = \frac{\sigma^4}{(n-1)^2}V\left(\left(\frac{n-1}{\sigma^2}\right)S^2\right) \\ &= \frac{2\sigma^4}{(n-1)} \end{aligned}$$

7
8
9
10
11 The variance of an estimator tells me about the
12 amount of error I should anticipate when
using that estimator.

SE = "standard error" of estimator = $\sqrt{\text{Variance}}$
SE of $s^2 = \frac{\sqrt{2}\sigma^2}{\sqrt{n-1}}$